QGIS WORKSHOP

KARTIKEYA GAUR

3rd year UG
CED

FEBURARY 2023








Teur
ਤੇਉਰ
Palheri
ਪਾਲ੍ਹੇਰੀ
ਮੁੱਲਾਪੁਰ
ਗਰੀਬਦਾਸ
ਪਰਤ੍ਹ
Karoran
Forest
Khudda
Lahora
Nayagaon
ਨਯਾਗਾਓ
Sukhna
Wildlife
Sanctuary
Jandpur
ਜੰਦਪੁਰ
Tohga
ਟੋਹਗਾ
Rajindra Park
Satish Law Agency
Rajipur
राजीपूर
Kuranwala
कुरंवाला
Kharar
ਖਰੜ
Taj Chandigarh
Modern hotel, dining
& a rooftop pool
Hotel Sunbeam
Modern rooms, plus
a restaurant & a bar
Sukteri
सुक्तेरी
Daun
ਦਾਊਂ
ISKCON Temple
Chandigarh
Garden
of Silence
Chandi Mandir
चांदी
मंदिर
Power Grid Substation
ਪਾਵਰ ਗਰਿੱਡ
ਸਬ ਸਟੇਸ਼ਨ
SECTOR 35
ਸੈਕਟਰ 35
MANSA DEVI
COMPLEX
मनसा देवी
कॉम्पलेक्स
GOVERNMENT
MEDICAL COLLEGE...
MANIMAJRA
Soorajpur
सूरजपुर
Sahibzada Ajit
Singh Nagar
ਸਾਹਿਬਜ਼ਾਦਾ
ਅਜੀਤ ਸਿੰਘ ਨਗਰ
Turquoise
Chic pick with a
revolving restaurant
DARIA
MAULI JAGRAN
Panchkula
पंचकुला
Nada Sahib
नाडा
साहिब
Landran
ਲਾਂਡਰਾਂ
Thunder Zone
ਥੰਡਰ ਜ਼ੋਨ ਜਲ ਅਤੇ
ਮਨੋਰੰਜਨ ਪਾਰਕ
Chandigarh
International
Airport
Mauli Baidwan
ਮੌਲੀ
ਬੈਦਵਾਨ
Rurka
Google
Bhabat
PEER MUCHALLA


zomato

LAYOUT

What we'll cover in this session



- Applications
- A bit of History
- Introduction to Remote Sensing
- GIS basics
- Quiz



APPLICATIONS

Still unexplored! Interestingly

HISTORY

Just a bit

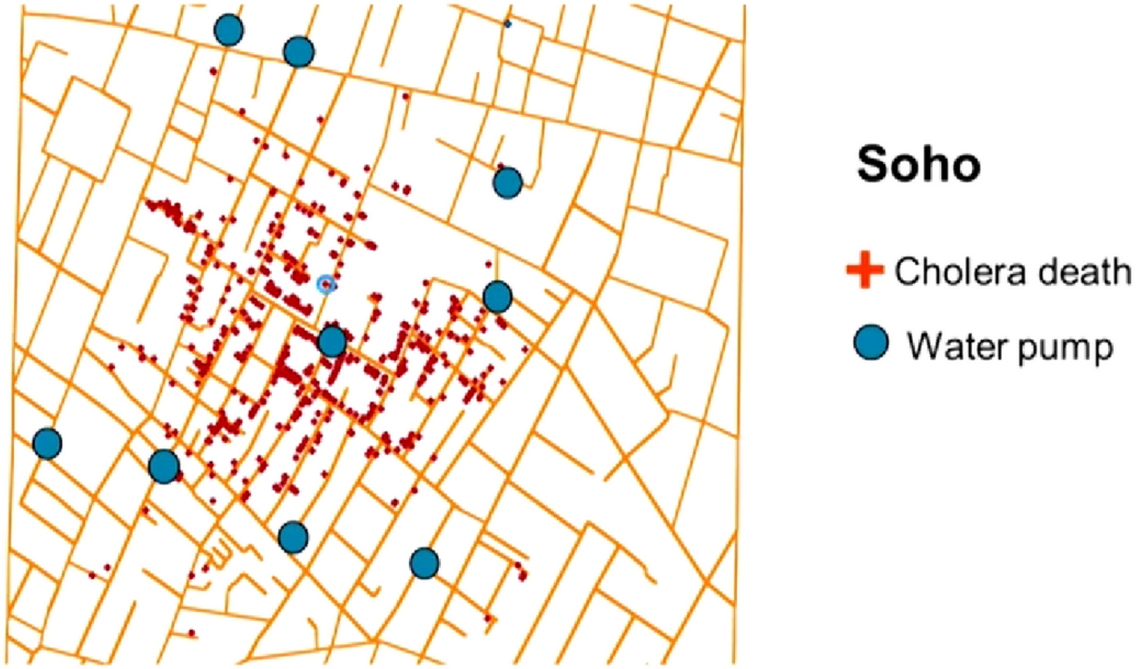
The field of geographic information systems (GIS) started in the 1960s as computers and early concepts of quantitative and computational geography emerged.

1963; Tomlinson created the design for automated computing to store and process large amounts of data, which enabled Canada to begin its national land-use management program. He also gave GIS its name.



GIS concepts are not new!

London cholera epidemic 1854



source: https://youtu.be/AyeANYXcsK4

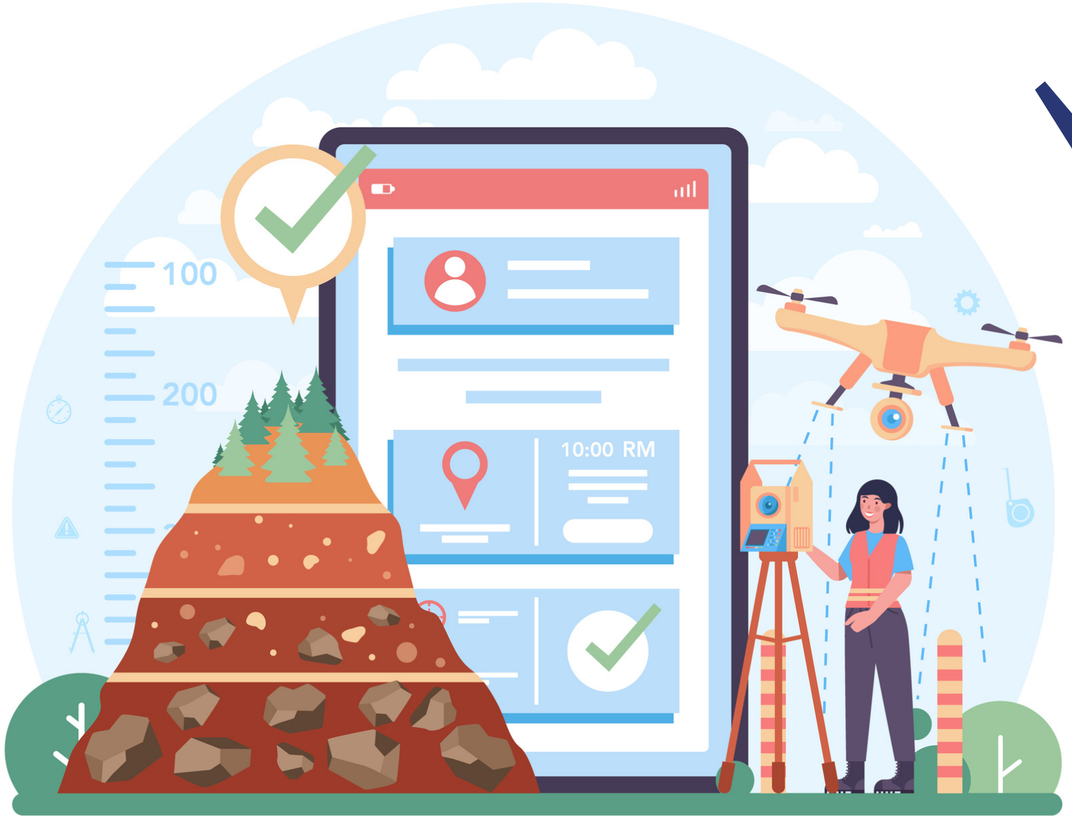
Remote sensing

Aditya L1

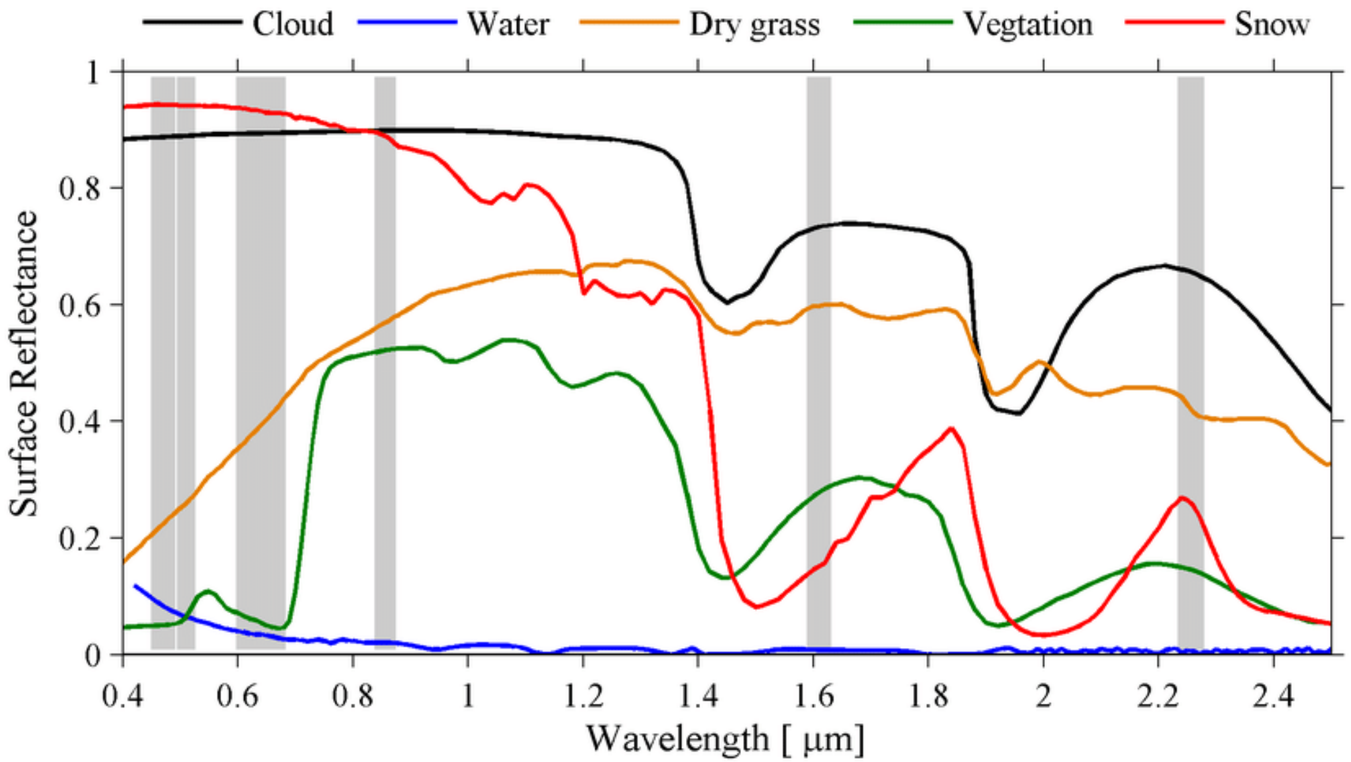
Going by the literal meaning, it means to sense something remotely, without touching it. As simple as that.



PLATFORMS







EMS and Spectral signatures

CGXUVIMR

Each object on earth behaves differently to the incidence of different wavelengths of EMS.
Amount of solar radiation.
Reflects more energy or less?

Remote Sensing and GIS

The ultimate nexus



- Data
- Information
- Analysis
- Modeling
- Desicion making

GIS basics

Geographic INFORMATION System

Just a software + capabilities to handle spatial data.

01

Definition

The real world is too complex for our immediate and direct understanding, so we create 'models' or abstractions of reality. Spatially referenced data act as a model of reality.

02

Spatial Relationships

Data Models

01

RASTER

Grid cells
Continous, generally.

02

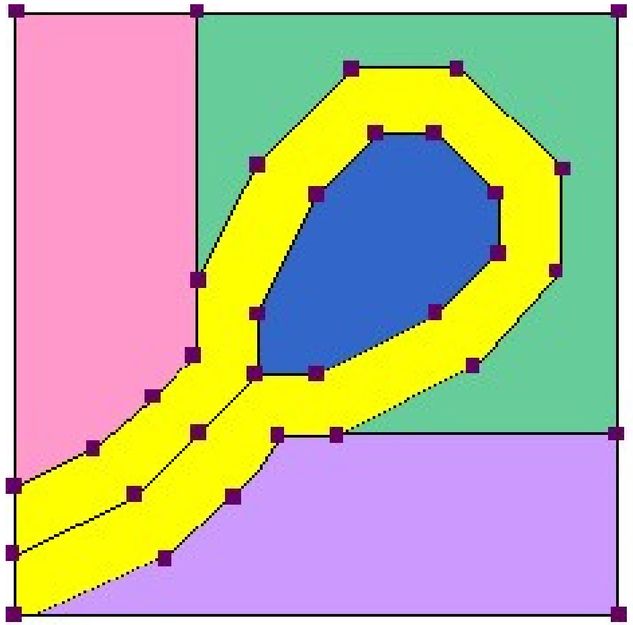
VECTOR

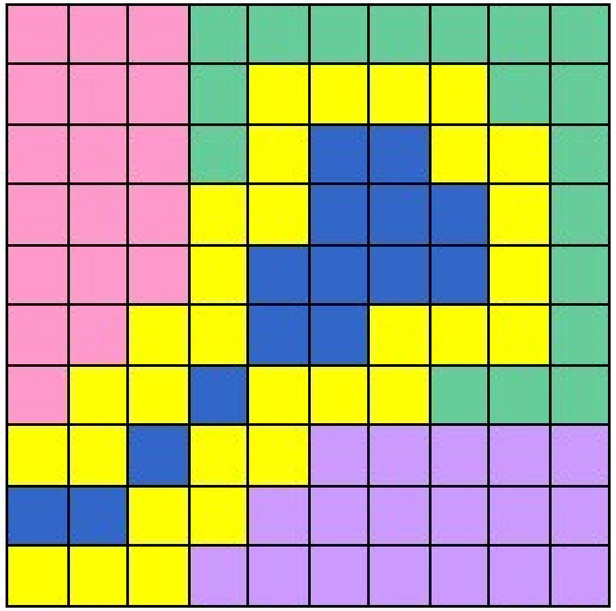
Points, line and polygon.
Discrete, generally.

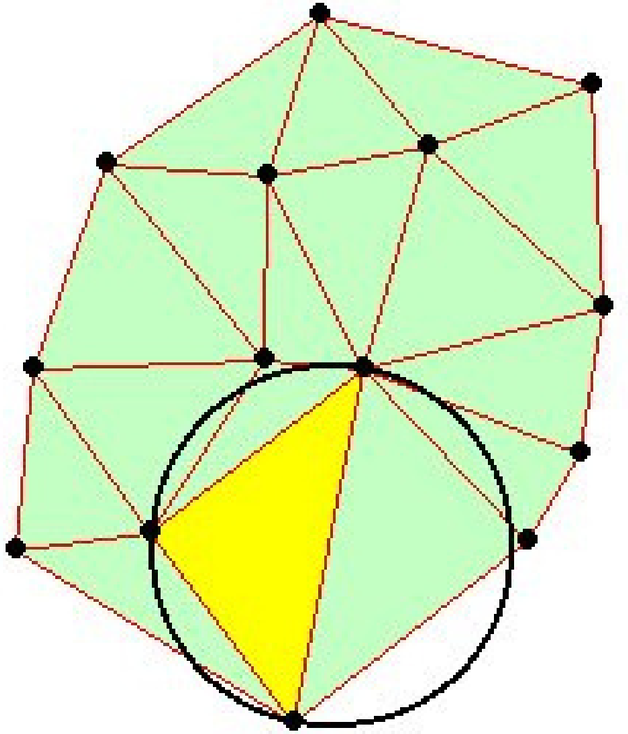
03

TIN

Triangulated Irregular Network
Delauney's triangulation.

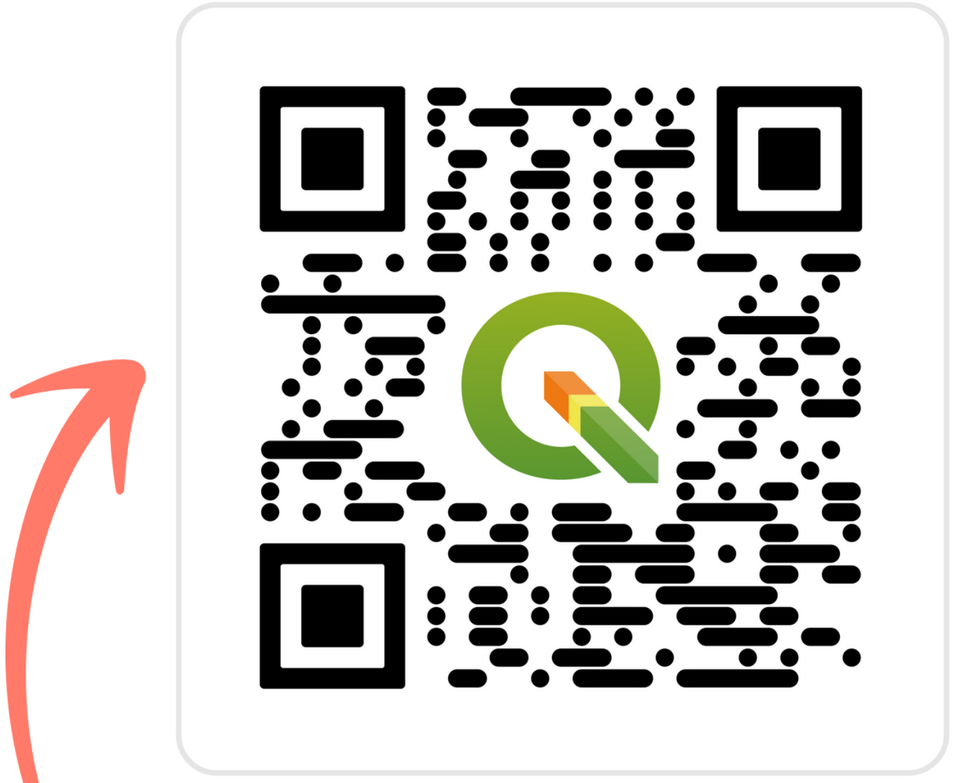








QUIZ



Scan Here



About the software

Open Source

- Some basics
- Coordinate system
- Excel to vector

MAPS, COORDINATE SYSTEM, AND PROJECTION

Curved lines to almost straight lines, through mathematical modeling.

Cyliinderical, Conic or Azimuthal?

A map represents the earth's surface on the plane surface with standards symbols.

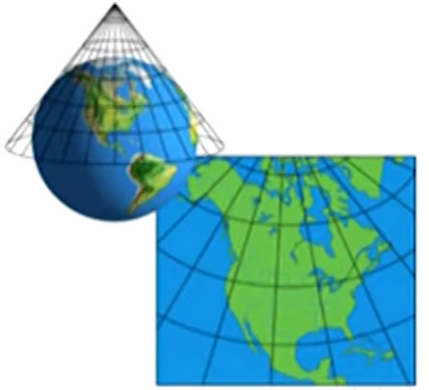
World Geodetic System 1984.

Choosing right projection is important.

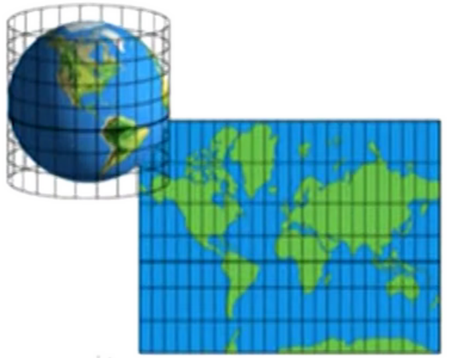
True shape and size.

Each country different extent and different projections.

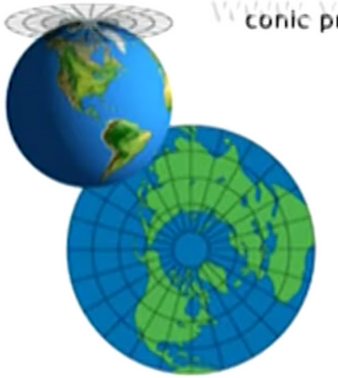
Map Projections



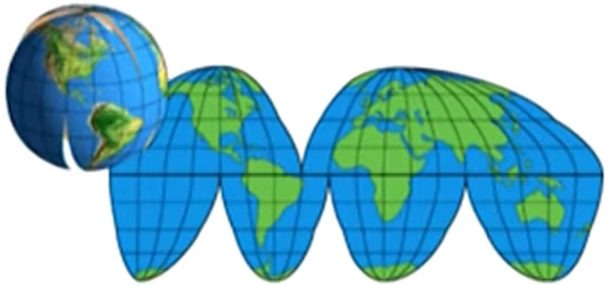
conic projection



cylindrical projection



plane projection



interrupted projection

source: https://youtu.be/vyWfUBhePz0

DID YOU KNOW?

5%

At any given time 5% of India’s population is on the move on its trains.

Indian Railways transport more than the entire population of **Australia** every day.

Railway Alignment

PARAMETERS

GRADIENT

Vertical and horizontal alignments are considered while planning, 1:150 is desired vertical alignment for broad gauge in plains whereas it could be up to 1:80 in hilly terrains. 1:150 refers to 1m rise vertically in 150m horizontal stretch. It is to ensure smooth movement of train and is standardised to keep everything in control.

CURVATURE

Since trains can’t take sharp turns we need to ensure that curves are always gradual in nature and don’t affect the speed of the train. Therefore, 6-degree curvature is acceptable but 4 degrees is the standard. Here, curvature refers to the angle made for arc length of 30m.



Feedback Form



THANK YOU